

Face Recognition using PCA (Eigenfaces) and ANN with Open-Set Recognition

Objective

Develop a face recognition system using Principal Component Analysis (PCA) for feature extraction and an Artificial Neural Network (ANN) for classification. Extend the system to perform open-set recognition by identifying unknown persons as 'Not Enrolled'.

Technologies

Python, OpenCV, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook.

Dataset

9 enrolled persons, 450 grayscale face images resized to 100x100 pixels. Training set: 270 images (60%). Testing set: 180 images (40%).

Methodology

Images were preprocessed (grayscale, resize, flatten), split into training and testing sets, transformed using PCA (Eigenfaces), and classified using an MLPClassifier (ANN). Multiple values of k were tested to select the best number of Eigenfaces. Open-set recognition was implemented by adding imposter images to the test set and applying a confidence threshold to classify low-confidence predictions as 'Not Enrolled'.

Results

Face Recognition Accuracy: approximately 75%. Known Person Accuracy (Open-set): 63.33%. Imposter Detection Rate: 24.24%.

Advantages

Dimensionality reduction using PCA, efficient ANN-based classification, and support for recognizing unknown persons.

Limitations

Performance depends on image quality and illumination. The confidence-threshold approach may still classify some unknown faces as enrolled users.

Future Work

Use deep-learning models such as FaceNet or ArcFace, improve unknown-person detection with distance-based methods, and develop a real-time recognition application.

Conclusion

The project successfully demonstrates face recognition using PCA and ANN and extends it to open-set recognition by detecting unknown individuals as 'Not Enrolled', fulfilling the internship project objectives.